

DBMS PROJECT

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Project Title: Pizza Sales Management System

INTRODUCTION:

Welcome to our Pizza Sales Management System database management system (DBMS) project. The design and implementation of a DBMS that can effectively manage pizza orders, menu details, and sales transactions is the aim of this project.

There is a growing need for efficient and reliable food order management systems as fast food services continue to expand rapidly. By developing a system that can track orders, item details, pricing, and pizza categories, our project aims to fulfil this demand.

The DBMS will include a structured database that captures all essential information related to the sales and operations of a pizza outlet. The system will feature a user-friendly interface and will be accessible to authorized staff. Its design ensures data accuracy, integrity, and efficiency in handling daily transactions.

TABLES:

- order_details

Columns:

- order_details_id
- order_id
- pizza_id
- quantity

- orders

Columns:

- order_id
- date
- time

- pizza_types

Columns:

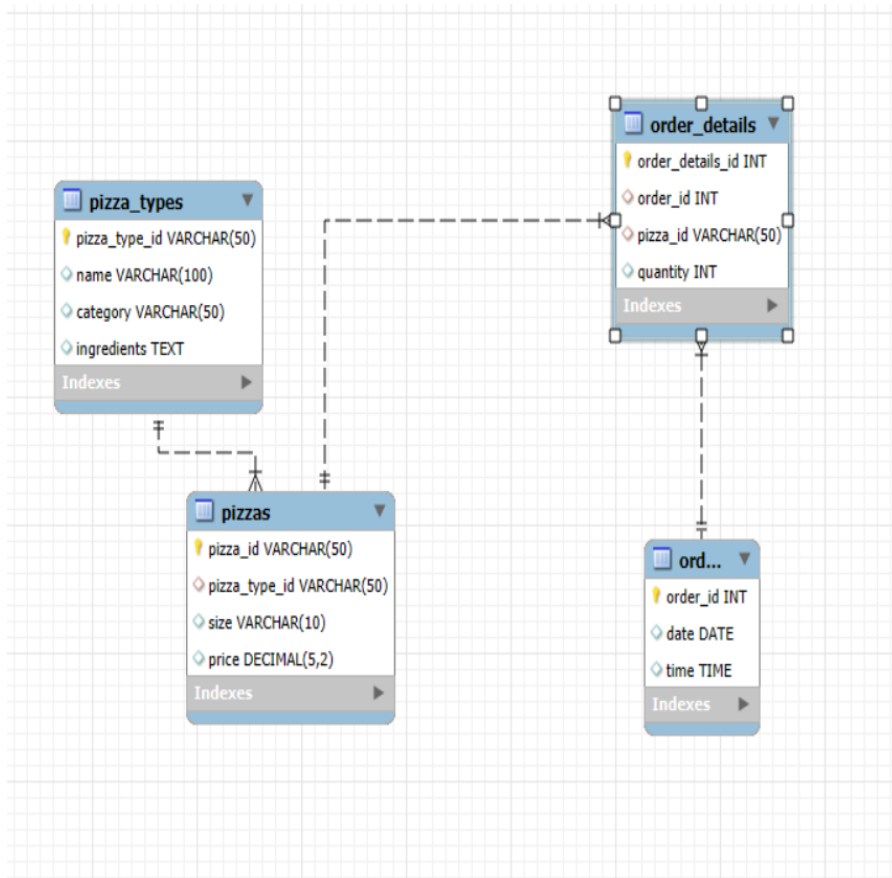
- pizza_type_id
- name
- category
- ingredients

- pizzas

Columns:

- pizza_id
- pizza_type_id
- size
- price

ER DIAGRAM:



RELATIONAL SCHEMA:

pizza_types: (pizza_type_id (PK), name, category, ingredients)

pizzas: (pizza_id (PK), pizza_type_id (FK), size, price)

orders: (order_id (PK), date, time)

order_details: (order_details_id (PK), order_id (FK), pizza_id (FK), quantity)

NORMALIZATION AND FUNCIONAL DEPENDENCIES:

The tables we created in this project are all in BCNF normal form:

orders Table:

orders (order_id, date, time)

Functional Dependencies:

- $\text{order_id} \rightarrow \text{date}$
- $\text{order_id} \rightarrow \text{time}$
- No Partial Dependencies exist. Hence, it is in **2NF**
- No Transitive Dependencies exist. Hence, it is in **3NF**

order_details Table:

order_details (order_details_id, order_id, pizza_id, quantity)

Functional Dependencies:

- $\text{order_details_id} \rightarrow \text{order_id}$
- $\text{order_details_id} \rightarrow \text{pizza_id}$
- $\text{order_details_id} \rightarrow \text{quantity}$
- No Partial Dependencies exist. Hence, it is in **2NF**
- No Transitive Dependencies exist. Hence, it is in **3NF**

pizzas Table:

pizzas (pizza_id, pizza_type_id, size, price)

Functional Dependencies:

- pizza_id → pizza_type_id
- pizza_id → size
- pizza_id → price
- No Partial Dependencies exist. Hence, it is in **2NF**
- No Transitive Dependencies exist. Hence, it is in **3NF**

pizza_types Table:

pizza_types (pizza_type_id, name, category, ingredients)

Functional Dependencies:

- pizza_type_id → name
- pizza_type_id → category
- pizza_type_id → ingredients
- No Partial Dependencies exist. Hence, it is in **2NF**
- No Transitive Dependencies exist. Hence, it is in **3NF**

Creating Database using SQL Developer:

orders Table:

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY,  
    date DATE,  
    time TIME  
);
```

pizza_types Table:

```
CREATE TABLE pizza_types (  
    pizza_type_id VARCHAR(50) PRIMARY KEY,  
    name VARCHAR(100),  
    category VARCHAR(50),  
    ingredients TEXT  
);
```

pizzas Table:

```
CREATE TABLE pizzas (  
    pizza_id VARCHAR(50) PRIMARY KEY,  
    pizza_type_id VARCHAR(50),  
    size VARCHAR(10),  
    price DECIMAL(5,2),  
    FOREIGN KEY (pizza_type_id) REFERENCES  
    pizza_types(pizza_type_id)  
);
```

order_details Table:

```
CREATE TABLE order_details (  
    order_details_id INT PRIMARY KEY,  
    order_id INT,  
    pizza_id VARCHAR(50),  
    quantity INT,  
    FOREIGN KEY (order_id) REFERENCES orders(order_id),  
    FOREIGN KEY (pizza_id) REFERENCES pizzas(pizza_id)  
);
```

Inserting Values In Database:

orders Table:

```
INSERT INTO orders (order_id, `date`, `time`) VALUES  
(1, '2025-07-01', '12:10:00'),  
(2, '2025-07-01', '12:45:00'),  
(3, '2025-07-02', '13:30:00'),  
(4, '2025-07-03', '14:15:00'),  
(5, '2025-07-03', '15:00:00');
```

order_id	date	time
1	2025-07-01	12:10:00
2	2025-07-01	12:45:00
3	2025-07-02	13:30:00
4	2025-07-03	14:15:00
5	2025-07-03	15:00:00

pizza_types Table:

```
I INSERT INTO pizza_types (pizza_type_id, name, category, ingredients) VALUES  
(  
'margherita', 'Margherita Pizza', 'Vegetarian', 'Tomato Sauce, Mozzarella,  
Basil'),  
  
(  
'pepperoni', 'Pepperoni Pizza', 'Classic', 'Tomato Sauce, Mozzarella,  
Pepperoni'),  
  
(  
'bbq_ckn', 'BBQ Chicken Pizza', 'Chicken', 'BBQ Sauce, Chicken, Red Onions,  
Mozzarella'),  
  
(  
'veggie_del', 'Veggie Delight Pizza', 'Vegetarian', 'Tomatoes, Olives, Bell  
Peppers, Onions, Mozzarella'),  
  
(  
'hawaiian', 'Hawaiian Pizza', 'Classic', 'Tomato Sauce, Ham, Pineapple,  
Mozzarella');
```

pizza_type_id	name	category	ingredients
bbq_ckn	BBQ Chicken Pizza	Chicken	BBQ Sauce, Chicken, Red Onions, Mozzarella
hawaiian	Hawaiian Pizza	Classic	Tomato Sauce, Ham, Pineapple, Mozzarella
margherita	Margherita Pizza	Vegetarian	Tomato Sauce, Mozzarella, Basil
pepperoni	Pepperoni Pizza	Classic	Tomato Sauce, Mozzarella, Pepperoni
veggie_del	Veggie Delight Pizza	Vegetarian	Tomatoes, Olives, Bell Peppers, Onions, Mozzarella

pizzas Table:

```
INSERT INTO pizzas (pizza_id, pizza_type_id, size, price) VALUES
('margherita_s', 'margherita', 'S', 8.99),
('margherita_m', 'margherita', 'M', 11.99),
('margherita_l', 'margherita', 'L', 14.99),
('pepperoni_s', 'pepperoni', 'S', 9.99),
('pepperoni_m', 'pepperoni', 'M', 12.99),
('pepperoni_l', 'pepperoni', 'L', 15.99),
('bbq_ckn_s', 'bbq_ckn', 'S', 10.49),
('bbq_ckn_m', 'bbq_ckn', 'M', 13.49),
('bbq_ckn_l', 'bbq_ckn', 'L', 16.49),
('veggie_del_s', 'veggie_del', 'S', 9.49),
('veggie_del_m', 'veggie_del', 'M', 12.49),
('veggie_del_l', 'veggie_del', 'L', 15.49),
('hawaiian_s', 'hawaiian', 'S', 9.75),
('hawaiian_m', 'hawaiian', 'M', 12.75),
('hawaiian_l', 'hawaiian', 'L', 15.75);
```

pizza_id	pizza_type_id	size	price
bbq_ckn_l	bbq_ckn	L	16.49
bbq_ckn_m	bbq_ckn	M	13.49
bbq_ckn_s	bbq_ckn	S	10.49
hawaiian_l	hawaiian	L	15.75
hawaiian_m	hawaiian	M	12.75
hawaiian_s	hawaiian	S	9.75
margherita_l	margherita	L	14.99
margherita_m	margherita	M	11.99
margherita_s	margherita	S	8.99
pepperoni_l	pepperoni	L	15.99
pepperoni_m	pepperoni	M	12.99
pepperoni_s	pepperoni	S	9.99
veggie_del_l	veggie_del	L	15.49
veggie_del_m	veggie_del	M	12.49
veggie_del_s	veggie_del	S	9.49

order_details Table:

```
INSERT INTO order_details (order_details_id, order_id, pizza_id, quantity)
VALUES
```

```
(1, 1, 'margherita_m', 1),
```

```
(2, 1, 'pepperoni_s', 2),
```

```
(3, 2, 'bbq_ckn_l', 1),
```

```
(4, 2, 'veggie_del_m', 1),
```

```
(5, 3, 'hawaiian_m', 2),
```

```
(6, 4, 'pepperoni_l', 1),
```

```
(7, 4, 'margherita_s', 1),
```

```
(8, 5, 'veggie_del_l', 1),
```

```
(9, 5, 'bbq_ckn_s', 2);
```

order_details_id	order_id	pizza_id	quantity
1	1	margherita_m	1
2	1	pepperoni_s	2
3	2	bbq_ckn_l	1
4	2	veggie_del_m	1
5	3	hawaiian_m	2
6	4	pepperoni_l	1
7	4	margherita_s	1
8	5	veggie_del_l	1
9	5	bbq_ckn_s	2

Queries:

Q1) Calculate the total revenue generated from pizza sales ?

ANS: -

```
select  
round(sum(order_details.quantity*pizzas.price),2) as total_sales  
from order_details  
join pizzas  
on order_details.pizza_id=pizzas.pizza_id;
```

	total_sales
▶	147.90

Q2) Join the necessary tables to find the total quantity of each pizza category ordered ?

ANS: -

```
select  
pizza_types.category,  
sum(order_details.quantity) as quantity  
from pizza_types  
join pizzas  
on pizza_types.pizza_type_id = pizzas.pizza_type_id  
join order_details  
on order_details.pizza_id = pizzas.pizza_id  
group by pizza_types.category  
order by quantity desc;
```

category	quantity
Classic	5
Vegetarian	4
Chicken	3

Q3) Determine the top 3 most ordered pizza types based on revenue?

ANS: -

```
select
pizza_types.name,
sum(order_details.quantity * pizzas.price) as revenue
from pizza_types
join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by revenue desc
limit 3;
```

name	revenue
BBQ Chicken Pizza	37.47
Pepperoni Pizza	35.97
Veggie Delight Pizza	27.98

Q4) Calculate the percentage contribution of each pizza type to total revenue?

ANS: -

```
select
    pizza_types.category,
    round(
        (sum(order_details.quantity * pizzas.price)) /
        (
            select
                sum(order_details.quantity * pizzas.price)
            from order_details
            join pizzas
                on order_details.pizza_id = pizzas.pizza_id
        ) * 100, 2
    ) as revenue
from pizza_types
join pizzas
    on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
    on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category
order by revenue desc;
```

category	revenue
Classic	41.56
Vegetarian	33.10
Chicken	25.33

Q5) Analyse the cumulative revenue generated over time?

ANS: -

```
select
    order_date,
    sum(revenue) over(order by order_date) as cum_revenue
from (
    select
        orders.date as order_date,
        sum(order_details.quantity * pizzas.price) as revenue
    from order_details
    join pizzas
        on order_details.pizza_id = pizzas.pizza_id
    join orders
        on orders.order_id = order_details.order_id
    group by orders.date
) as sales;
```

order_date	cum_revenue
2025-07-01	60.95
2025-07-02	86.45
2025-07-03	147.90